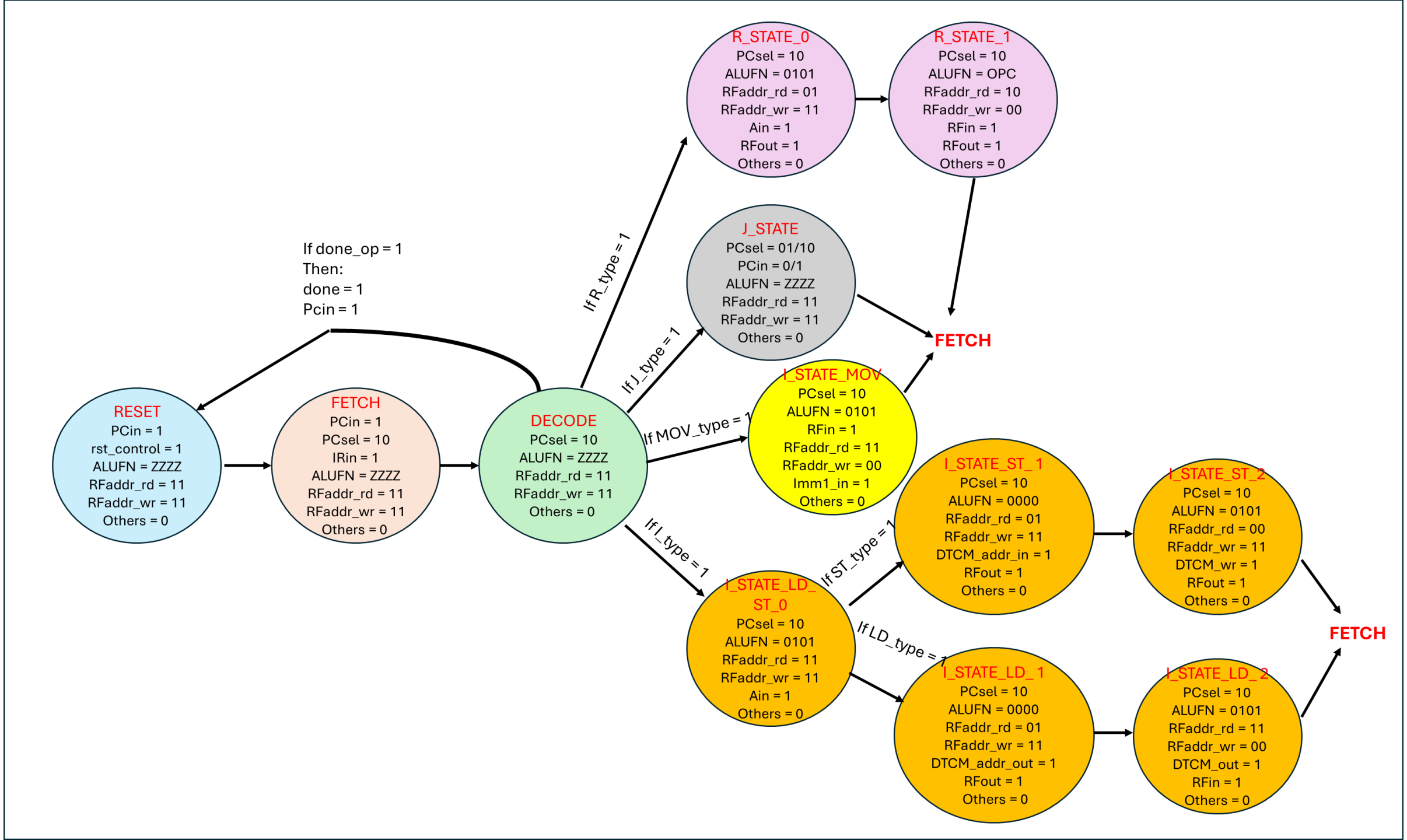
RESET
PCin = 1
rst_control = 1
ALUFN = ZZZZ
RFaddr_rd = 11
RFaddr_wr = 11
Others = 0
FETCH
PCin = 1
PCsel = 10
IRin = 1
ALUFN = ZZZZ
RFaddr_rd = 11
RFaddr_wr = 11
Others = 0
DECODE
PCsel = 10
ALUFN = ZZZZ
RFaddr_rd = 11
RFaddr_wr = 11
Others = 0
If done_op = 1
Then:
done = 1
Pcin = 1
If R_type = 1
R_STATE_0
PCsel = 10
ALUFN = 0101
RFaddr_rd = 01
RFaddr_wr = 11
Ain = 1
RFout = 1
Others = 0
R_STATE_1
PCsel = 10
ALUFN = OPC
RFaddr_rd = 10
RFaddr_wr = 00
RFin = 1
RFout = 1
Others = 0
If J_type = 1
J_STATE
PCsel = 01/10
PCin = 0/1
ALUFN = ZZZZ
RFaddr_rd = 11
RFaddr_wr = 11
Others = 0
FETCH
If MOV_type = 1
I_STATE_MOV
PCsel = 10
ALUFN = 0101
RFin = 1
RFaddr_rd = 11
RFaddr_wr = 00
Imm1_in = 1
Others = 0
If I_type = 1
I_STATE_LD_ST_0
PCsel = 10
ALUFN = 0101
RFaddr_rd = 11
RFaddr_wr = 11
Ain = 1
Others = 0
If ST_type = 1
I_STATE_ST_ 1
PCsel = 10
ALUFN = 0000
RFaddr_rd = 01
RFaddr_wr = 11
DTCM_addr_in = 1
RFout = 1
Others = 0
I_STATE_ST_2
PCsel = 10
ALUFN = 0101
RFaddr_rd = 00
RFaddr_wr = 11
DTCM_wr = 1
RFout = 1
Others = 0
If LD_type = 1
I_STATE_LD_ 1
PCsel = 10
ALUFN = 0000
RFaddr_rd = 01
RFaddr_wr = 11
DTCM_addr_out = 1
RFout = 1
Others = 0
I_STATE_LD_ 2
PCsel = 10
ALUFN = 0101
RFaddr_rd = 11
RFaddr_wr = 00
DTCM_out = 1
RFin = 1
Others = 0
FETCH